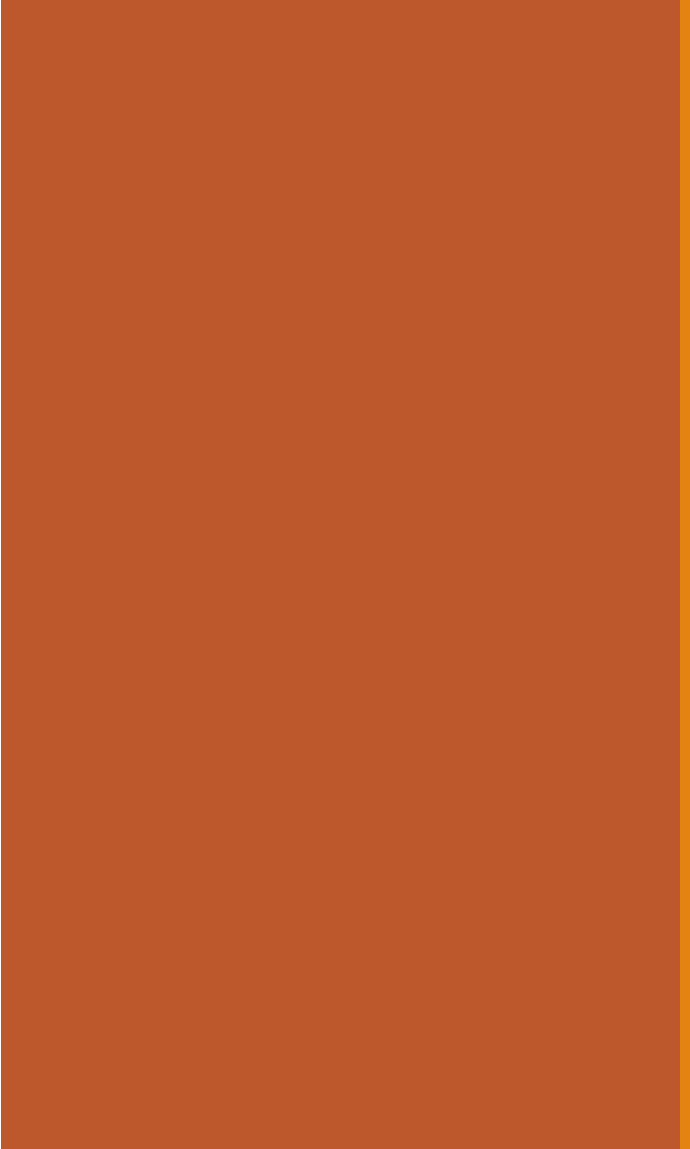


18: Central Limit Theorem

Jerry Cain
May 10th, 2024

[Lecture Discussion on Ed](#)



iid Random Variables

Independence of multiple random variables

Review

We have independence of n discrete random variables $X_1, X_2, \dots, X_n$ if for all $x_1, x_2, \dots, x_n$:

$$P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n) = \prod_{i=1}^n P(X_i = x_i)$$
$$p_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n p_{X_i}(x_i)$$

We have independence of n continuous random variables $X_1, X_2, \dots, X_n$ if for all $x_1, x_2, \dots, x_n$:

$$P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) = \prod_{i=1}^n P(X_i \leq x_i)$$
$$f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) = \prod_{i=1}^n f_{X_i}(x_i)$$

i.i.d. random variables

Consider n variables $X_1, X_2, \dots, X_n$.

this part is new and wasn't required in the last slide.

$X_1, X_2, \dots, X_n$ are **independent** and **identically distributed** if

- $X_1, X_2, \dots, X_n$ are independent, and
- All have the same PMF (if discrete) or PDF (if continuous).

$\Rightarrow E[X_i] = \mu$ for $i = 1, \dots, n$

$\Rightarrow \text{Var}(X_i) = \sigma^2$ for $i = 1, \dots, n$

*in fact, all centralized moments would be equal if truly IID.
so $E[(X_i - \mu)^3]$ are the same
 $E[(X_i - \mu)^3]$ are the same, and so forth.*

Same thing:

i.i.d.

iid

IID

Quick check

Are $X_1, X_2, \dots, X_n$ iid with the following distributions?

1. $X_i \sim \text{Exp}(\lambda)$, X_i independent
2. $X_i \sim \text{Exp}(\lambda_i)$, X_i independent
3. $X_i \sim \text{Exp}(\lambda)$, $X_1 = X_2 = \dots = X_n$
4. $X_i \sim \text{Bin}(n_i, p)$, X_i independent

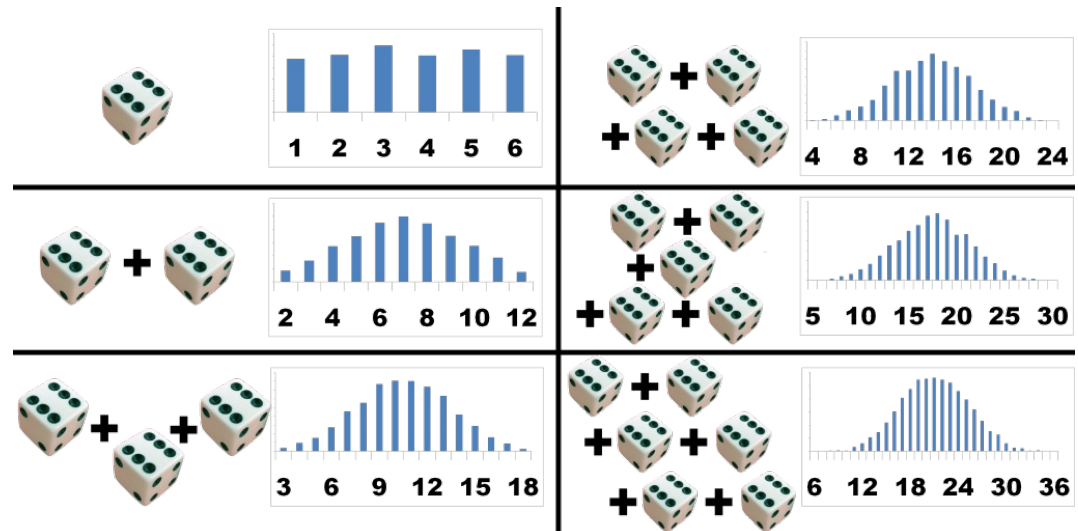


Quick check

Are $X_1, X_2, \dots, X_n$ iid with the following distributions?

1. $X_i \sim \text{Exp}(\lambda)$, X_i independent 
2. $X_i \sim \text{Exp}(\lambda_i)$, X_i independent  (unless λ_i equal)
3. $X_i \sim \text{Exp}(\lambda)$, $X_1 = X_2 = \dots = X_n$  dependent: $X_1 = X_2 = \dots = X_n$
4. $X_i \sim \text{Bin}(n_i, p)$, X_i independent  (unless n_i equal)
Note underlying Bernoulli RVs are iid!

Central Limit Theorem



[source](#)

Central Limit Theorem

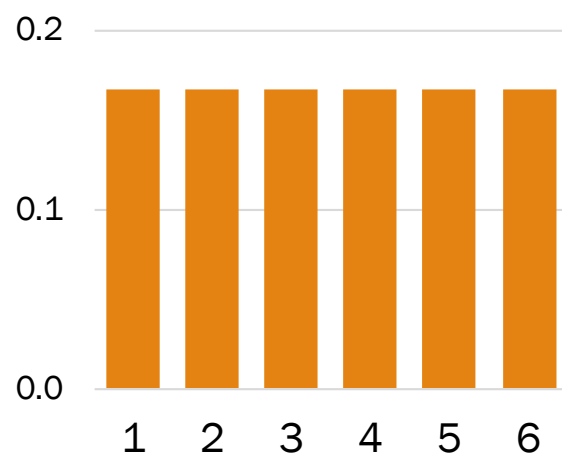
Consider n **independent and identically distributed (iid)** variables $X_1, X_2, \dots, X_n$ with $E[X_i] = \mu$ and $\text{Var}(X_i) = \sigma^2$.

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2) \quad \text{As } n \rightarrow \infty$$

The sum of n **iid** random variables is normally distributed with mean $n\mu$ and variance $n\sigma^2$.
and standard deviation $\sqrt{n}\sigma$.

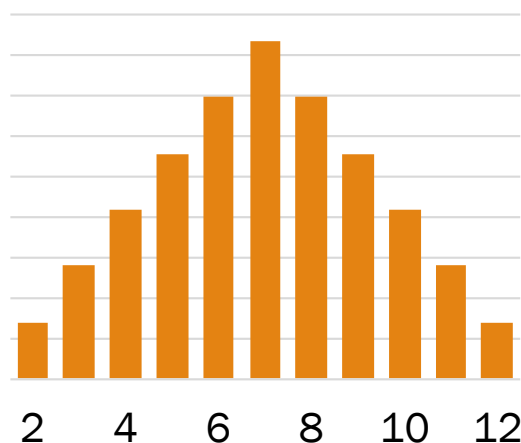
Sum of dice rolls

Roll n independent dice. Let X_i be the outcome of roll i . X_i are iid



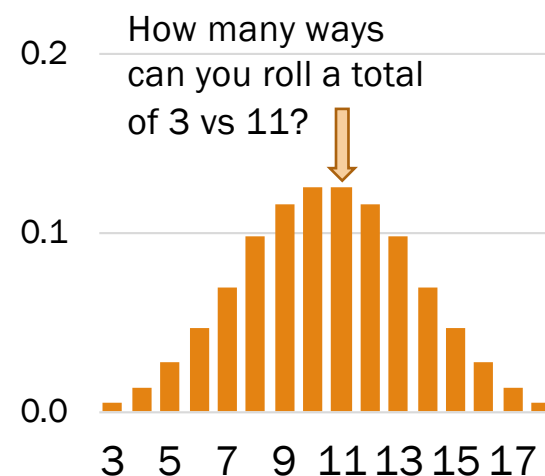
$$\sum_{i=1}^1 X_i$$

Sum of 1 die roll



$$\sum_{i=1}^2 X_i$$

Sum of 2 dice rolls



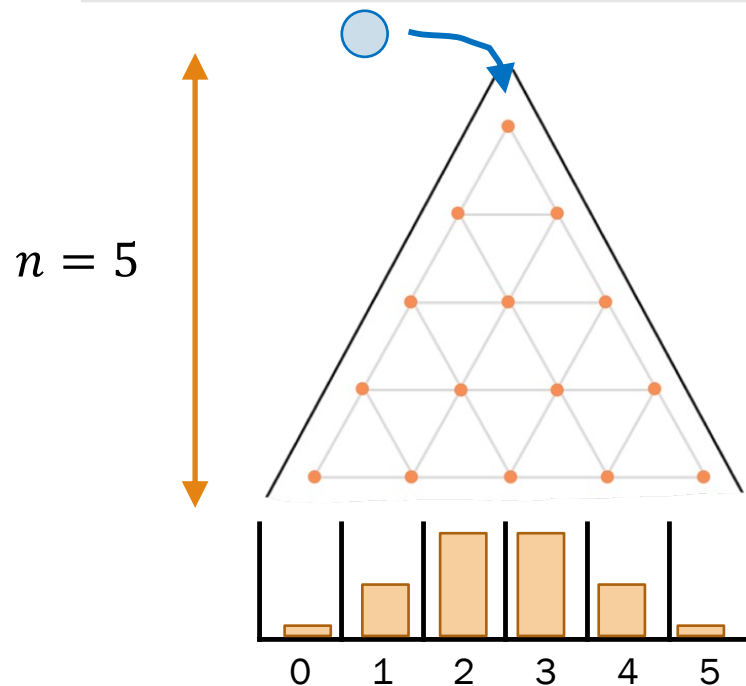
$$\sum_{i=1}^3 X_i$$

Sum of 3 dice rolls

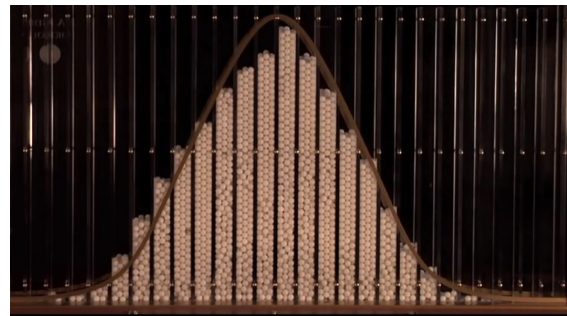
CLT explains a lot

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2) \quad \text{As } n \rightarrow \infty$$

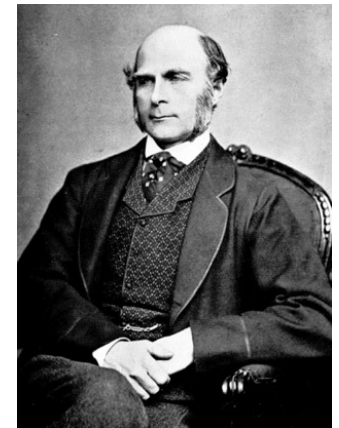
The sum of n iid random variables is normally distributed with mean $n\mu$ and variance $n\sigma^2$.



*the sum of lots of iid
Bernoulli is
distributed as a Gaussian.*



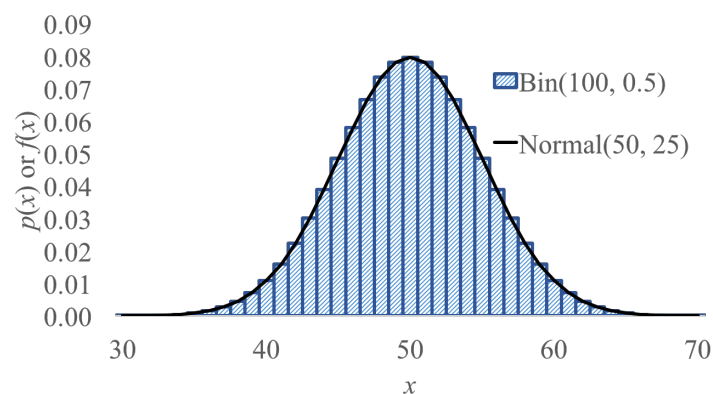
Galton Board, by Sir Francis Galton
(1822-1911)



CLT explains a lot

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2) \quad \text{As } n \rightarrow \infty$$

The sum of n **iid** random variables is normally distributed with mean $n\mu$ and variance $n\sigma^2$.



Normal approximation of Binomial

Sum of iid Bernoulli RVs $\approx$ Normal

Proof:

Let $X_i \sim \text{Ber}(p)$ for $i = 1, \dots, n$, where X_i are iid
 $E[X_i] = p$, $\text{Var}(X_i) = p(1 - p)$

$$X = \sum_{i=1}^n X_i$$

*POV: forget about Binomials!
the sum of lots of iid Bernoullis is
well-approximated
by a Gaussian.* (X ~ Bin(n, p))

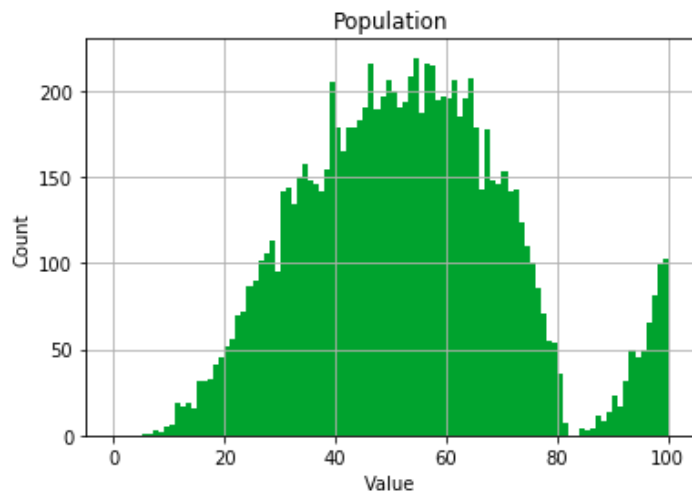
$$X \sim \mathcal{N}(n\mu, n\sigma^2) \quad (\text{CLT, as } n \rightarrow \infty)$$

$$X \sim \mathcal{N}(np, np(1 - p)) \quad (\text{substitute mean, variance of Bernoulli})$$

CLT explains a lot

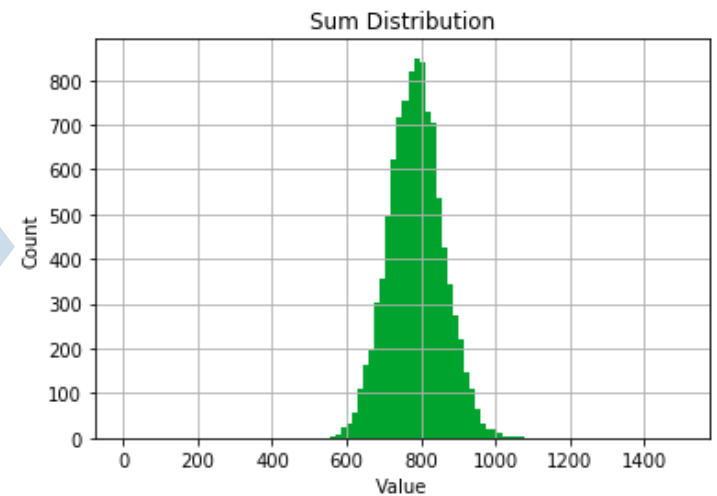
$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2) \quad \text{As } n \rightarrow \infty$$

The sum of n iid random variables is normally distributed with mean $n\mu$ and variance $n\sigma^2$.



Distribution of X_i

Sample of
size 15,
sum values



Distribution of $\sum_{i=1}^{15} X_i$

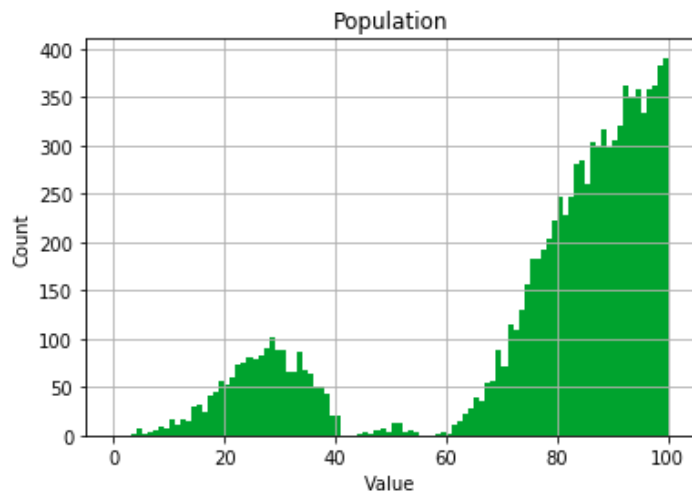
CLT explains a lot

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2) \quad \text{As } n \rightarrow \infty$$

The sum of n iid random variables is normally distributed with mean $n\mu$ and variance $n\sigma^2$.

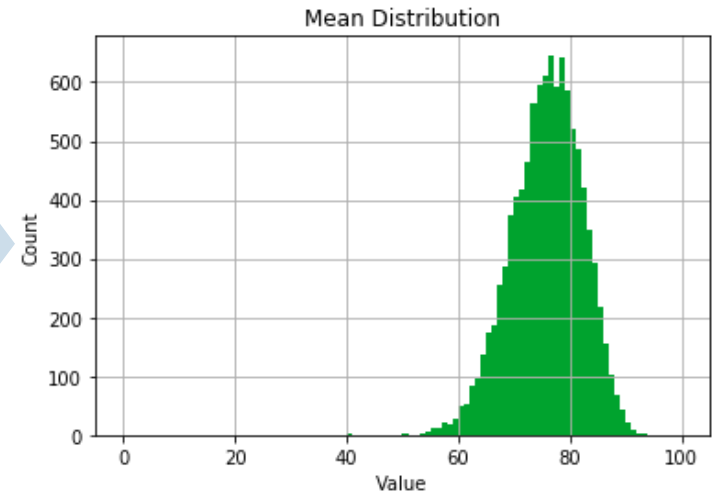
extension of above statement:

if the sum of n iid variables is a Gaussian, that that sum divided by n is a Gaussian as well.



Distribution of X_i

Sample of
size 15,
average values



Distribution of $\frac{1}{15} \sum_{i=1}^{15} X_i$

Proof of CLT

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2) \quad \text{As } n \rightarrow \infty$$

The sum of n **iid** random variables is normally distributed with mean $n\mu$ and variance $n\sigma^2$.

Sketch of Proof:

- The Fourier Transform of a PDF is its **characteristic function**.
- Take the characteristic function of the probability mass of the sample distance from the mean, divided by standard deviation
- Show that this approaches an exponential function in the limit as $n \rightarrow \infty$: $f(x) = e^{-\frac{x^2}{2}}$
- This function just happens to be the characteristic function of the Standard Normal, $Z \sim \mathcal{N}(0,1)$.

(the full [proof](#) is beyond the scope of CS109)

Sum of n independent Uniform RVs

Let $X = \sum_{i=1}^n X_i$ be sum of **iid** RVs, where $X_i \sim \text{Uni}(0,1)$. $\mu = E[X_i] = 1/2$
 $\sigma^2 = \text{Var}(X_i) = 1/12$

For different n , how close is the CLT approximation of $P(X \leq n/3)$?

$n = 2$:

Exact

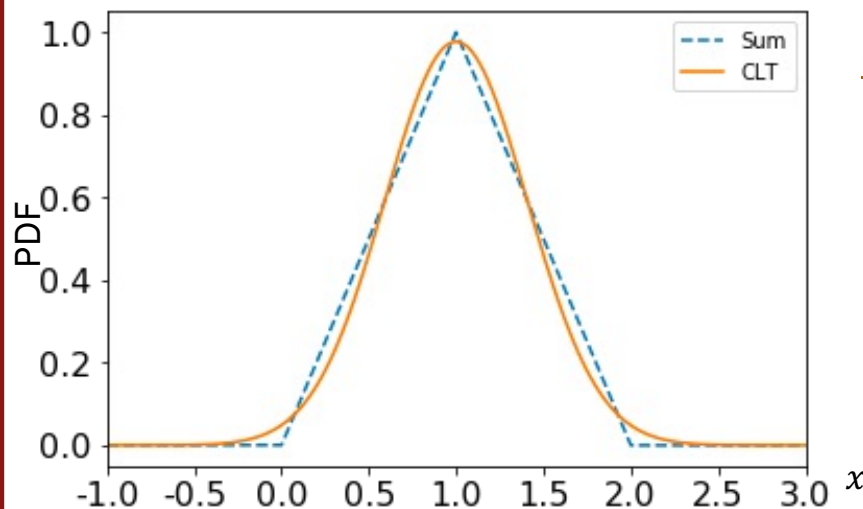
$$P(X \leq 2/3) \approx 0.2222$$

CLT approximation

$$X \approx Y \sim \mathcal{N}(n\mu, n\sigma^2) \Rightarrow Y \sim \mathcal{N}(1, 1/6)$$

$$P(X \leq 2/3) \approx P(Y \leq 2/3)$$

$$= \Phi\left(\frac{2/3 - 1}{\sqrt{1/6}}\right) \approx 0.2071$$



Sum of n independent Uniform RVs

Let $X = \sum_{i=1}^n X_i$ be sum of **iid** RVs, where $X_i \sim \text{Uni}(0,1)$. $\mu = E[X_i] = 1/2$
 $\sigma^2 = \text{Var}(X_i) = 1/12$

For different n , how close is the CLT approximation of $P(X \leq n/3)$?

$n = 5$:

Exact

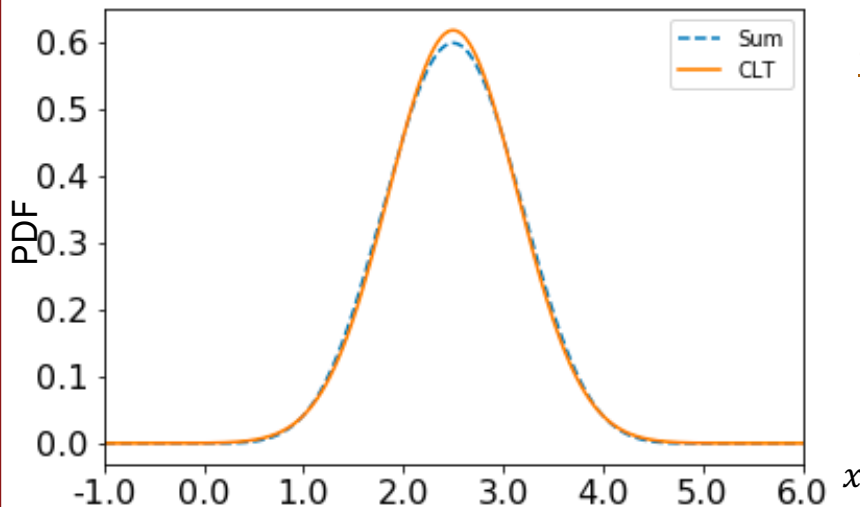
$$P(X \leq 5/3) \approx 0.1017$$

CLT approximation

$$X \approx Y \sim \mathcal{N}(n\mu, n\sigma^2) \Rightarrow Y \sim \mathcal{N}(5 \cdot \frac{1}{2}, 5 \cdot \frac{1}{12}) \Rightarrow Y \sim \mathcal{N}(5/2, 5/12)$$

$$P(X \leq 5/3) \approx P(Y \leq 5/3)$$

$$= \Phi\left(\frac{5/3 - 5/2}{\sqrt{5/12}}\right) \approx 0.0984$$



Sum of n independent Uniform RVs

Let $X = \sum_{i=1}^n X_i$ be sum of **iid** RVs, where $X_i \sim \text{Uni}(0,1)$. $\mu = E[X_i] = 1/2$
 $\sigma^2 = \text{Var}(X_i) = 1/12$

For different n , how close is the CLT approximation of $P(X \leq n/3)$?

$n = 10$:

Exact

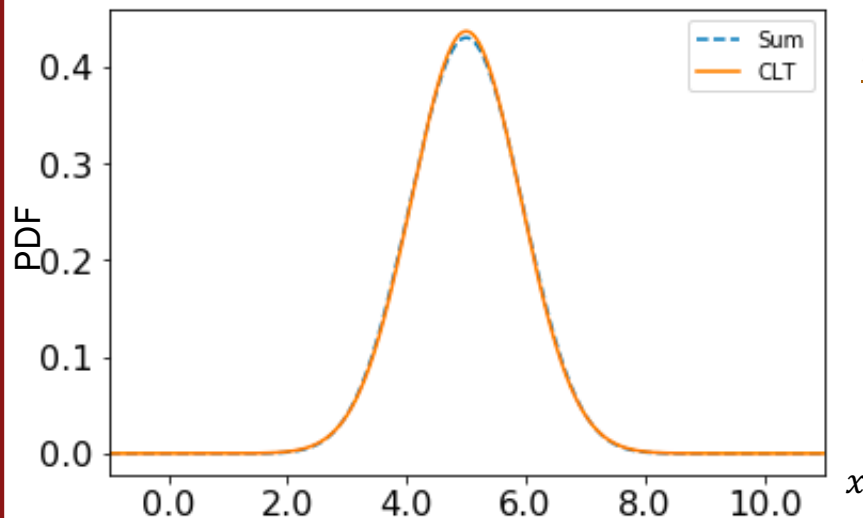
$$P(X \leq 10/3) \approx 0.0337$$

CLT approximation

$$X \approx Y \sim \mathcal{N}(n\mu, n\sigma^2) \Rightarrow Y \sim \mathcal{N}(5, 5/6)$$

$$P(X \leq 10/3) \approx P(Y \leq 10/3)$$

$$= \Phi\left(\frac{10/3 - 5}{\sqrt{5/6}}\right) \approx 0.0339$$

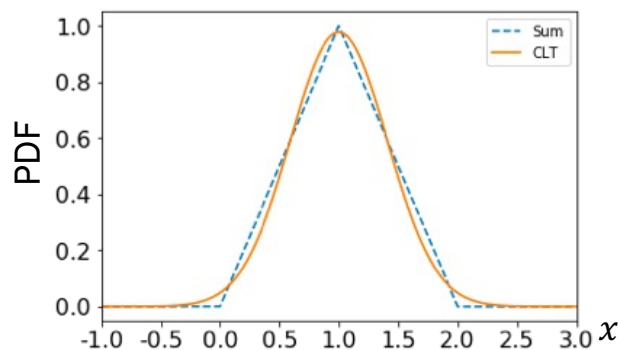


Sum of n independent Uniform RVs

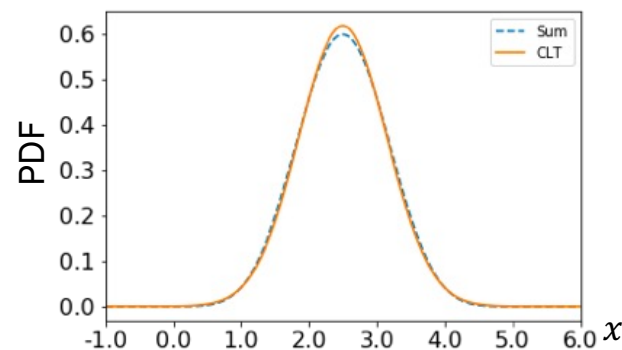
Let $X = \sum_{i=1}^n X_i$ be sum of iid RVs, where $X_i \sim \text{Uni}(0,1)$. $\mu = E[X_i] = 1/2$
 $\sigma^2 = \text{Var}(X_i) = 1/12$

For different n , how close is the CLT approximation of $P(X \leq n/3)$?

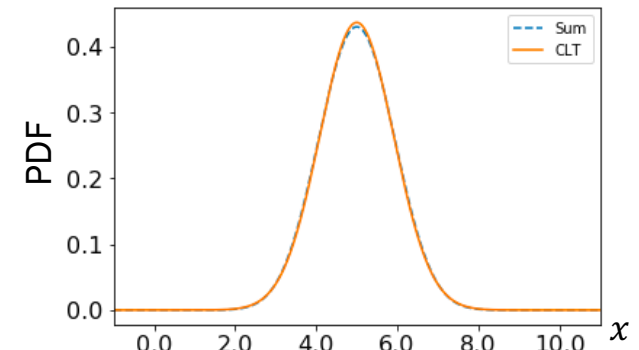
$n = 2$:



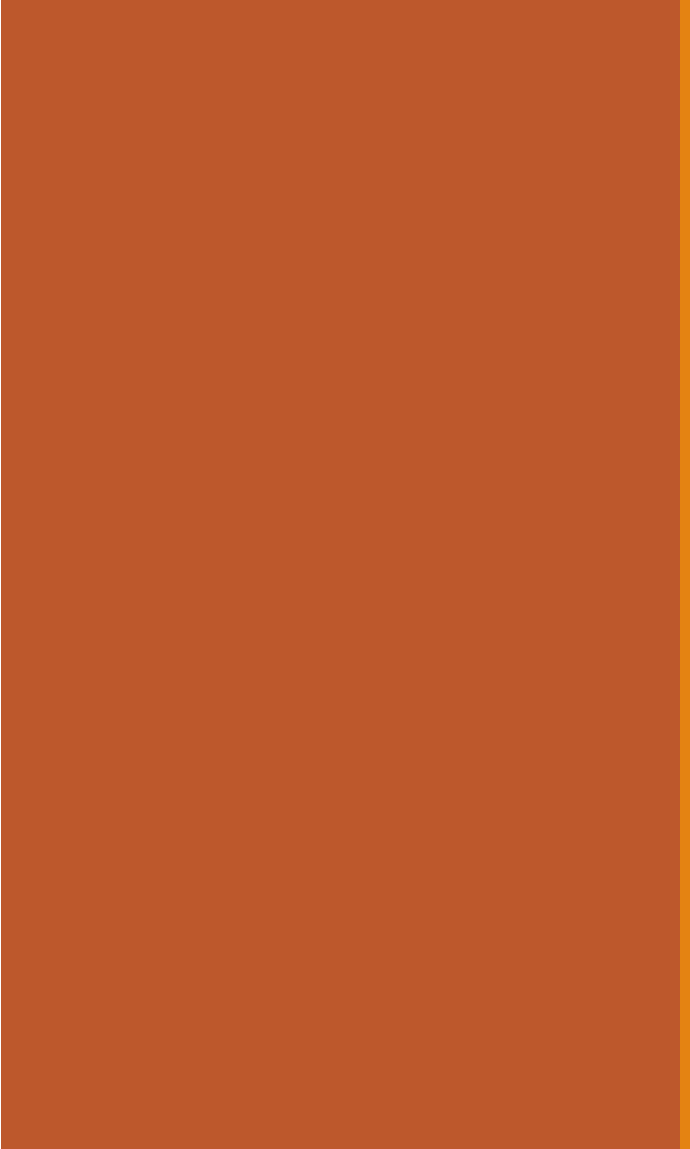
$n = 5$:



$n = 10$:



Most books will tell you that CLT holds if $n \geq 30$, but it might hold for even smaller n depending on the distribution of your iid X_i 's, particularly if they're symmetric.



Sample Statistics

What about other functions?

Let $X_1, X_2, \dots, X_n$ be iid, where $E[X_i] = \mu, \text{Var}(X_i) = \sigma^2$. As $n \rightarrow \infty$:

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

Sum of iid RVs

?

Average of iid RVs
(sample mean)

?

Max of iid RVs

What about other functions?

Let $X_1, X_2, \dots, X_n$ be iid, where $E[X_i] = \mu, \text{Var}(X_i) = \sigma^2$. As $n \rightarrow \infty$:

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

Sum of iid RVs

?

Average of iid RVs
(sample mean)

?

Max of iid RVs

Distribution of sample mean

Let $X_1, X_2, \dots, X_n$ be **iid**, where $E[X_i] = \mu, \text{Var}(X_i) = \sigma^2$. As $n \rightarrow \infty$:

Define: $\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$ (sample mean) $Y = \sum_{i=1}^n X_i$ (sum)

$Y \sim \mathcal{N}(n\mu, n\sigma^2)$ (CLT, as $n \rightarrow \infty$)

$$\bar{X} = \frac{1}{n} Y$$

$$\begin{aligned} E[\bar{X}] &= \frac{1}{n} E[Y] = \mu \\ \text{Var}(\bar{X}) &= \frac{1}{n^2} \text{Var}(Y) = \frac{n\sigma^2}{n^2} = \frac{\sigma^2}{n} \end{aligned}$$

$$\bar{X} \sim \mathcal{N}(\text{?}, \text{?})$$

$$\left(\frac{n\mu}{n}, \frac{n\sigma^2}{n^2} \right) \Rightarrow \left(\mu, \frac{\sigma^2}{n} \right)$$

(Linear transform of a Normal)

$$\frac{1}{n} \sum_{i=1}^n X_i \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

The average of **iid** random variables (i.e., **sample mean**) is normally distributed with mean μ and variance σ^2/n .

Demo: http://onlinestatbook.com/stat_sim/sampling_dist/

Lisa Yan, Chris Piech, Mehran Sahami, and Jerry Cain, CS109, Spring 2024

What about other functions?

Let $X_1, X_2, \dots, X_n$ be iid, where $E[X_i] = \mu, \text{Var}(X_i) = \sigma^2$. As $n \rightarrow \infty$:

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

Sum of iid RVs

$$\frac{1}{n} \sum_{i=1}^n X_i \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

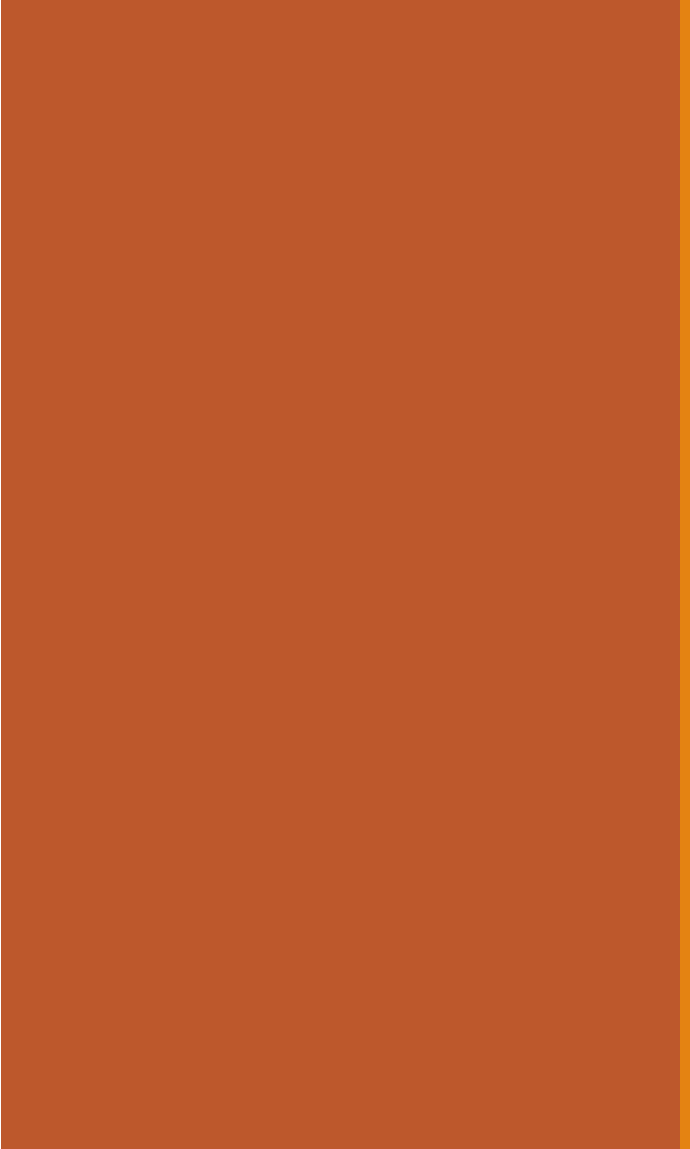
Average of iid RVs
(sample mean)

Gumbel

Max of iid RVs

} to be clear, CLT
has little to say
about maximums.

(see Fisher-Tippett Gnedenko Theorem)



Exercises

Dice game

$$\text{As } n \rightarrow \infty: \sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

You will roll 10 6-sided dice ($X_1, X_2, \dots, X_{10}$).

- Let $X = X_1 + X_2 + \dots + X_{10}$, the total value of all 10 rolls.
- You win if $X \leq 25$ or $X \geq 45$.



[To the demo!](#)



Dice game

$$\text{As } n \rightarrow \infty: \sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

You will roll 10 6-sided dice($X_1, X_2, \dots, X_{10}$).

- Let $X = X_1 + X_2 + \dots + X_{10}$, the total value of all 10 rolls.
- You win if $X \leq 25$ or $X \geq 45$.



And now the truth (according to the CLT)...

1. Define RVs and state goal.

$$E[X_i] = 3.5, \\ \text{Var}(X_i) = 35/12$$

Want: $P(X \leq 25 \text{ or } X \geq 45)$

Approximate:

?

2. Solve.



Dice game

$$\text{As } n \rightarrow \infty: \sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

You will roll 10 6-sided dice($X_1, X_2, \dots, X_{10}$).

- Let $X = X_1 + X_2 + \dots + X_{10}$, the total value of all 10 rolls.
- You win if $X \leq 25$ or $X \geq 45$.



And now the truth (according to the CLT)...

1. Define RVs and state goal.

$$E[X_i] = 3.5, \\ \text{Var}(X_i) = 35/12$$

Want: $P(X \leq 25 \text{ or } X \geq 45)$

Approximate:

$$X \approx Y \sim \mathcal{N}(10(3.5), 10(35/12))$$

2. Solve.

$$P(Y \leq 25.5) + P(Y \geq 44.5) \quad \text{or} \quad 1 - P(25.5 \leq Y \leq 44.5)$$

Dice game

$$\text{As } n \rightarrow \infty: \sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

You will roll 10 6-sided dice($X_1, X_2, \dots, X_{10}$).

- Let $X = X_1 + X_2 + \dots + X_{10}$, the total value of all 10 rolls.
- You win if $X \leq 25$ or $X \geq 45$.



And now the truth (according to the CLT)...

1. Define RVs and state goal.

$$E[X_i] = 3.5, \\ \text{Var}(X_i) = 35/12$$

Want: $P(X \leq 25 \text{ or } X \geq 45)$

Approximate:

$$X \approx Y \sim \mathcal{N}(10(3.5), 10(35/12))$$

2. Solve.

$$P(Y \leq 25.5) + P(Y \geq 44.5) = \Phi\left(\frac{25.5 - 35}{\sqrt{10(35/12)}}\right) + \left(1 - \Phi\left(\frac{44.5 - 35}{\sqrt{10(35/12)}}\right)\right)$$

$$\approx \Phi(-1.76) + (1 - \Phi(1.76)) \approx (1 - 0.9608) + (1 - 0.9608) = \mathbf{0.0786}$$

Dice game

$$\text{As } n \rightarrow \infty: \sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

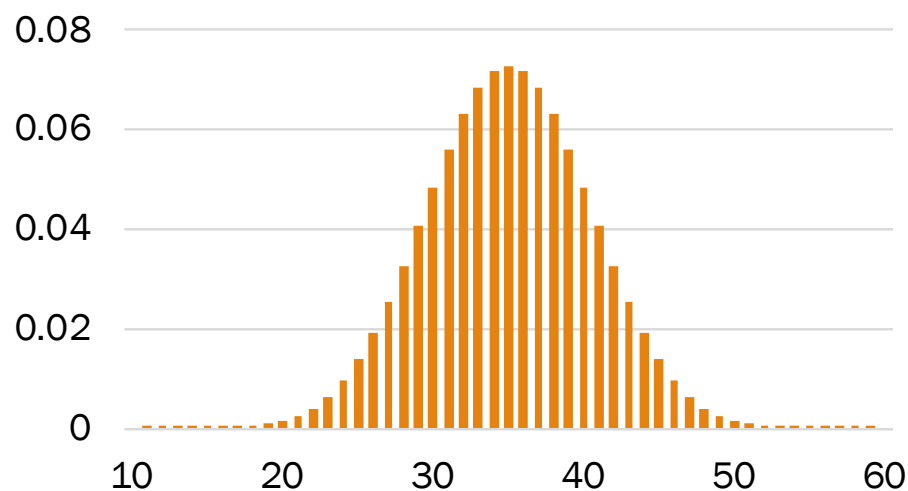
You will roll 10 6-sided dice($X_1, X_2, \dots, X_{10}$).

- Let $X = X_1 + X_2 + \dots + X_{10}$, the total value of all 10 rolls.
- You win if $X \leq 25$ or $X \geq 45$.



And now the truth (according to the CLT)...

Check out
the [code!](#)



(via CLT)

$$\approx P(Y \leq 25.5) + P(Y \geq 44.5) \\ \approx 0.0786$$

(exact, by computer)

$$P(X \leq 25 \text{ or } X \geq 45) = 0.0780$$

(sampling via computer)

$$P(X \leq 25 \text{ or } X \geq 45) \approx 0.0776$$

Summary: Working with the CLT

Let $X_1, X_2, \dots, X_n$ **iid**, where $E[X_i] = \mu, \text{Var}(X_i) = \sigma^2$. As $n \rightarrow \infty$:

$$\sum_{i=1}^n X_i \sim \mathcal{N}(n\mu, n\sigma^2)$$

Sum of **iid** RVs

$$\frac{1}{n} \sum_{i=1}^n X_i \sim \mathcal{N}\left(\mu, \frac{\sigma^2}{n}\right)$$

Average of **iid** RVs
(sample mean)



If X_i is discrete:
Use the **continuity correction** on Y !

Next time

Central Limit Theorem:

- Sample mean $\bar{X} \sim \mathcal{N}(\mu, \sigma^2/n)$
- If we know μ and σ^2 , we can compute probabilities on sample mean $\bar{X}$ of a given sample size n

In real life:

- Yes, the CLT still holds. It always holds!
- But we **often don't know** μ or σ^2 of our original distribution
- However, we can collect data (a sample of size n)
- How can we **estimate** the values μ and σ^2 from our sample? And how reliable are those estimates?

...until next time!